
OpenAffect-EEG: Resource-Matched Evaluation of EEG Added Value

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Abstract

1 EEG affect systems may benefit from labelled user calibration and repeated stimuli
2 as well as EEG. A higher system score therefore does not, by itself, establish
3 incremental value from EEG. We introduce OpenAffect-EEG, an evaluation method
4 that fixes test trials, varies participant-calibration and repeated-stimulus label doses,
5 preserves population-training size, and compares EEG predictors with EEG-free
6 priors receiving the same labels. A complementary execution-contract checker
7 tests observable resource parity, data lineage, and deployment claims. In a bounded
8 sensitivity analysis on EmoEEG-MC and urban-image appraisal, two settings
9 each of band-power regression, frozen LaBraM, and standard EEGNet cover
10 complete dose grids. Label resources consistently improve the EEG-free prior,
11 whereas all mean matched EEG-increment intervals include zero. This is not
12 evidence of equivalence or absent EEG information: it limits attribution of total
13 system gains. Executed paths from two pinned external toolkits illustrate why split
14 names and trial-intersection checks do not by themselves certify broader evaluation
15 contracts. We retain earlier auditory, condition-level, and representation audits,
16 including uncertain results. The contribution is a reproducible comparison object
17 and checkable evidence for its assumptions, not a new encoder or a latent-emotion
18 decomposition.

1 Introduction

19 An EEG emotion system can use more than the current EEG recording. It may have ratings from
20 a target user’s previous trials, ratings of the same stimulus from other users, or source-released
21 normative stimulus scores. These resources are legitimate when available at deployment. The
22 evaluation problem arises when an EEG system receives them but its EEG-free comparator does
23 not, or when a protocol name conceals their availability. Canonical affect datasets contain repeated
24 stimuli and individual self-reports, making this distinction concrete [7, 12, 30].

25 Existing research establishes the importance of trial-level leakage prevention, participant and stimulus
26 holdout, target-user calibration, and identity audits [3–5, 15, 19]. Our question is narrower: *on the*
27 *same test trials, what changes when the EEG and EEG-free predictors are given the same participant*
28 *and stimulus label resources?* A protocol can exclude every test trial and still leave this comparison
29 unmatched. Conversely, resource matching alone does not establish generalization to identities that
30 have been observed during calibration.

31 We distinguish three score differences: total system gain against an uncalibrated prior, calibration gain
32 available without EEG, and the remaining EEG difference against that calibrated comparator. The
33 point is not that subtraction is new, or that more labels can help. It is that the comparison requires an
34 explicit, executable resource design: fixed test support, controlled population-training size, identical
35 calibration labels, and declared external priors. The same design also specifies which deployment
36 claims can be checked. We make three contributions:
37

Table 1: Complementary evaluation objects. This is a scoped comparison of published purposes, not an exhaustive absence-of-feature claim.

Approach	Question addressed	Additional object evaluated here
Trial/identity holdout	Do forbidden trials or identities cross partitions?	Dose-controlled resources with a fixed test and population size.
EEGAIN [13]	How do models compare in a standardized pipeline?	Does an EEG gain persist against a comparator with the same calibration labels?
FMScope [19]	What identity structure is encoded, and how does erasure affect decoding?	What labelled resources supported this prediction contrast and deployment claim?
OpenAffect-EEG	What can be attributed to EEG under a declared resource contract?	Paired score contrasts plus verifiable resource and execution records.

1. **A resource-matched evaluation object.** A fixed-support response surface independently doses target-participant calibration and other-participant ratings of test stimuli, with training-row replacement and paired EEG-free controls. We define its estimand and assumptions rather than claiming causal identity isolation.
2. **Evidence about attribution under controlled resources.** Complete grids and bounded training sensitivity show how favourable total system gains can coexist with uncertain matched EEG increments. Earlier models, auditory results, and unsupported comparisons remain available instead of being replaced by a favourable subset.
3. **Reusable checks of observable execution contracts.** Trial lineage, predictor resources, preprocessing and selection roles, and fixed test hashes produce allow, block, or unverifiable decisions. Pinned external code paths and separately identified synthetic faults document what these checks do and do not establish.

2 Relation to Existing Evaluation Work

EEGAIN standardizes datasets, preprocessing, models, splits, metrics, and prediction logging [13]. FMScope audits frozen representations with subject-axis, spectral, layer, and within-subject diagnostics [19]. Both address genuine problems; neither is a weak baseline to be defeated by adding more features to a checker. Table 1 locates our contribution by the question and observable inputs, not by an unsupported claim that no related method exists.

Calibration, joint person–stimulus holdout, and EEG toolboxes are established [3, 17, 22]. Likewise, identity probes, linear erasure, data cards, and provenance hashes are not claimed as individually novel [1, 9, 19, 26]. Outside EEG, supervision-conditioned benchmarking, controlled metric stress tests, and documented benchmark criteria are also established [10, 21, 27]. Dose curves, simulation, and checklists are therefore not separate novelty claims here. Our scope is resource-conditioned evaluation of observed self-reports. It is not the physiological mechanism audited by FMScope, nor a claim to exceed the dataset coverage or classification results of EEGAIN.

3 Resource-Matched Evaluation

3.1 Trial graph and resource contract

Each trial has a participant identity i , stimulus identity j , observed report y , context, and source lineage. Windows inherit their complete trial’s partition. Let T_b be the fixed test trials in identity block b and $D_b^{p,s}$ the population development data at doses (p, s) . Let $C_{i,b}^p$ contain p labelled calibration trials from target participant i , on stimuli outside the test stimulus set. Repeated-stimulus dose s counts population ratings per test stimulus from other participants. We record a contract

$$\mathcal{R}_b(p, s) = (T_b, D_b^{p,s}, \{C_{i,b}^p\}_i, N_b, A_b, F_b, V_b), \quad (1)$$

where N_b is the fixed population-training row count, A_b lists additional label resources such as normative scores, and F_b, V_b record preprocessing fit and model-selection trials. This notation describes actual supplied resources; it is not an identity-information measure.

73 Repeated-stimulus rows replace context- and target-matched training donors, so population row
 74 counts remain fixed. Calibration rows do not fit the population model; they estimate a post-fit
 75 participant mean residual offset. Dose sets are deterministic and nested. Removing donors changes
 76 training composition; we control count and approximate label/context balance, not every property of
 77 the training distribution. Nor is total labelled-data budget fixed across the participant axis: additional
 78 calibration is the resource being measured.

79 The principal invariants are: no test trial in fit, selection, or calibration; identical test trials across cells
 80 and models; exact realised doses; no target participant in population fitting; and identical non-EEG
 81 labels for paired predictors. Test identities are initially jointly held out, but at nonzero doses the
 82 corresponding identities have been exposed. Released normative scores, where available, remain
 83 separately declared at zero stimulus-label dose. Zero dose therefore does not generally mean no
 84 stimulus knowledge. Unverifiable session, device, or upstream pretraining identities cannot support a
 85 strict isolation certificate.

86 3.2 Predictors and the estimand

87 The population prior P uses fit-set stimulus means, falling back to declared source normative ratings
 88 or the fit-global mean. PC adds the target user’s mean calibration residual without EEG. E adds an
 89 EEG-predicted population residual to P , and EC calibrates E using exactly the same trial labels as
 90 PC . Both calibrated predictors use the same mean-offset rule. The residual is arithmetic relative to
 91 the specified prior, not an identified private emotion. Direct EEG prediction is retained as a secondary
 92 control.

93 For score functional Q on the same T_b , define

$$G_{\text{total}}(p, s) = Q(EC) - Q(P), \quad (2)$$

$$G_{\text{cal}}(p, s) = Q(PC) - Q(P), \quad (3)$$

$$\Delta_{\text{EEG}}(p, s) = Q(EC) - Q(PC). \quad (4)$$

94 Thus $G_{\text{total}} = G_{\text{cal}} + \Delta_{\text{EEG}}$ exactly. This is a telescoping score identity, not an additive decomposi-
 95 tion of CCC into neural mechanisms. G_{total} can assess an entire system package; it cannot isolate
 96 EEG added value when calibration resources differ. At $p = 0$, $PC = P$ and the matched contrast
 97 reduces to $Q(E) - Q(P)$.

98 Our primary summary is the uniform mean of Δ_{EEG} in CCC over five fixed identity blocks and the
 99 complete grid $\{0, 1, 2, 4, 8\}^2$. We report paired MAE differences as a secondary metric; positive
 100 MAE differences mean greater error. Uniform grid weighting is a specified benchmark utility, not an
 101 estimated deployment-frequency distribution. Endpoint tables illustrate comparison choice but do not
 102 replace this primary summary. Every grid cell is retained in the aggregate release.

103 3.3 Observable execution checks

104 The checker consumes a trial-identity table, an execution record, and boolean deployment require-
 105 ments. Unlike a split-name lookup, it compares executed UID sets for population labels, calibra-
 106 tion labels, test support, preprocessing, and model selection. Required missing evidence yields
 107 unverifiable, not a pass. A detected violation yields block; otherwise supplied observable ev-
 108 idence may allow the requested checks. Source identity authenticity and upstream pretraining
 109 exposure remain unverifiable even when observable checks pass.

110 For example, a calibrated target-user system can pass test-trial isolation yet fail a requested unseen-
 111 participant contract. Two predictors can also share every train/test split but fail resource parity
 112 because only one receives calibration labels. These are different checks with different inputs. A
 113 basic intersection checker is not expected to infer undeclared resource use. The resource checker
 114 complements it; it is not a universally more accurate detector.

115 4 Experimental Evidence

116 4.1 Coverage and bounded training sensitivity

117 The latest coverage analysis uses EmoEEG-MC (30 eligible participants, 42 stimuli; valence and
 118 arousal) and Urban Appraisal (62 participants, 56 street images; genuine valence trials only) [30, 32].

Table 2: Changing the comparison at the predeclared maximum dose $(p, s) = (8, 8)$. Total system gain $EC - P$ includes the EEG-free calibration gain $PC - P$. The last column is the paired percentile interval for $EC - PC$, not a difference of interval endpoints. These descriptive endpoints do not replace the primary uniform-grid mean.

Task	Setting	$EC - P$	$PC - P$	$EC - PC$	Paired interval
EmoEEG-MC	Band-power R1	+0.1312	+0.1420	-0.0108	[-0.0429, +0.0219]
EmoEEG-MC	Band-power R2	+0.1318	+0.1420	-0.0102	[-0.0403, +0.0206]
EmoEEG-MC	LaBraM R1	+0.0961	+0.1420	-0.0459	[-0.1098, +0.0139]
EmoEEG-MC	LaBraM R2	+0.1325	+0.1420	-0.0095	[-0.0419, +0.0243]
EmoEEG-MC	EEGNet L1	+0.1400	+0.1420	-0.0020	[-0.0065, +0.0027]
EmoEEG-MC	EEGNet L2	+0.1420	+0.1420	0.0000	[-0.0040, +0.0043]
Urban	Band-power R1	+0.0639	+0.0774	-0.0135	[-0.0561, +0.0362]
Urban	Band-power R2	+0.0643	+0.0774	-0.0131	[-0.0398, +0.0199]
Urban	LaBraM R1	+0.0587	+0.0774	-0.0186	[-0.0563, +0.0126]
Urban	LaBraM R2	+0.0689	+0.0774	-0.0085	[-0.0373, +0.0172]
Urban	EEGNet L1	+0.0885	+0.0774	+0.0111	[-0.0147, +0.0360]
Urban	EEGNet L2	+0.0851	+0.0774	+0.0078	[-0.0064, +0.0252]

119 Deterministic participant and stimulus blocks test every eligible identity once, producing 252 and
120 695 test trials respectively. Only diagonal identity blocks are tested, not all possible pairs. Urban is
121 an external appraisal task, not an identical replication of the emotion-induction paradigm. These data
122 have been used in earlier analyses; the new blocks are not an untouched confirmatory cohort.

123 Each task includes band-power Ridge, frozen LaBraM, and the standard Braindecode EEGNet
124 [11, 14], each with two bounded settings, across five blocks and all 25 cells: 1,500 model-cell results
125 in total, not independent observations. Ridge settings vary the candidate regularization grid; EEGNet
126 settings vary learning rate. Validation selects hyperparameters and training duration; population
127 refitting combines training and validation, never test or calibration trials. The direct and residual
128 EEGNet branches have 200 recorded selection/refit learning histories. They are finite and nonconstant,
129 which checks execution but does not establish optimal training. Full settings, coverage, preprocessing,
130 and historical model panels appear in the Appendix.

131 4.2 Paired uncertainty

132 Crossed bootstrap applies the same participant and stimulus multiplicities to all paired predictors,
133 cells, and blocks. Of 2,000 draws, 1,985 are valid for EmoEEG-MC and 2,000 for Urban. We
134 report percentile intervals and retain an explicitly heuristic small-cluster t-normal sensitivity analysis.
135 Intervals condition on the fixed fitted predictions; they do not include full retraining uncertainty.
136 Fixed blocks and settings are not independent study replications. Small-cluster coverage diagnostics
137 remain in the release, and nominal coverage is not guaranteed. We distinguish pointwise intervals
138 from separately supplied dose-family simultaneous bands; neither corrects all model comparisons.

139 **Stress-testing the evaluator.** Before interpreting observed data, we pass the same matched-contrast
140 analyzer through prespecified Gaussian crossed simulations with analytic population CCC. Across
141 16 trial-signal settings (100 replicates each), positive-detection rates are 0.85–1.00 and pointwise
142 coverage is 0.92–1.00. A labels-only predictor gives exact zero, and no positive detection is observed
143 in 16 independent-noise settings; the latter is finite simulation evidence, not a zero false-positive
144 rate. Fresh 300-replicate diagnostics give 0.933 percentile coverage in the previous 32-participant/24-
145 stimulus problem case, but the minimum falls to 0.837 with 8 participants and 6 stimuli. We therefore
146 validate the comparison’s intended behavior while retaining small-support interval failure as a limit;
147 Appendix N.5 reports all scenario roles and methods.

148 4.3 What the matched comparator changes

149 Table 2 retains every setting at the same predeclared maximum dose. For EmoEEG-MC LaBraM R1,
150 total gain relative to P is +0.0961, while the EEG-free calibration gain is +0.1420. The matched
151 EEG difference is -0.0459 with interval $[-0.1098, +0.0139]$. In Urban, the same setting has total
152 gain +0.0587, EEG-free calibration gain +0.0774, and matched difference -0.0186 with interval

Table 3: Primary uniform-grid mean matched EEG increment in CCC, with paired percentile intervals. Each task averages five fixed identity blocks and all 25 dose cells. Settings are sensitivity checks, not independent replications.

Setting	EmoEEG-MC: mean [interval]	Urban: mean [interval]
Band-power R1	-0.0069 [-0.0333, +0.0221]	-0.0060 [-0.0432, +0.0376]
Band-power R2	-0.0091 [-0.0306, +0.0145]	-0.0069 [-0.0312, +0.0216]
LaBraM R1	-0.0320 [-0.0676, +0.0104]	-0.0173 [-0.0482, +0.0112]
LaBraM R2	-0.0137 [-0.0376, +0.0139]	-0.0035 [-0.0238, +0.0150]
EEGNet L1	-0.0001 [-0.0040, +0.0047]	+0.0073 [-0.0056, +0.0217]
EEGNet L2	+0.0003 [-0.0028, +0.0040]	+0.0086 [-0.0043, +0.0215]

153 [-0.0563, +0.0126]. Thus a positive system-level point gain is not evidence for a positive EEG-
 154 specific contrast. The latter comparisons remain uncertain; we do not claim a statistically established
 155 universal reversal or absence of EEG information.

156 The full-grid primary summaries in Table 3 have intervals including zero for all 12 task–setting
 157 combinations under both interval methods. No equivalence margin was prespecified. These findings
 158 therefore limit claims of a stable average positive increment in the evaluated settings; they do not
 159 establish equivalence or a ceiling on future EEG methods. The secondary MAE contrast for LaBraM
 160 R1 is positive in both tasks under both interval constructions, indicating increased error for that
 161 configuration; other configurations include uncertain and interval-sensitive effects. All CCC and
 162 MAE settings, rather than selected significant rows, are reported in Appendix N.

163 The EEG-free prior is itself resource-sensitive. From zero doses, increasing only participant calibration
 164 to eight trials changes PC CCC by +0.2036 in EmoEEG-MC and +0.1564 in Urban; changing
 165 only stimulus-label dose to eight changes it by +0.4907 and +0.3408. The corresponding percentile
 166 and t-normal intervals are positive. This is not novel evidence that labels help. It explains why
 167 a non-EEG resource comparator is necessary when interpreting the much smaller matched EEG
 168 contrasts.

169 4.4 Earlier evidence remains part of the record

170 The original EmoEEG-MC and MusicEEG surfaces, standard versus compact EEGNet, CBraMod,
 171 LaBraM, same-architecture random controls, frozen EMOD, and shrinkage-calibration analyses
 172 remain in Appendix C.2. Their support and protocols are distinct from the latest coverage blocks
 173 and must not be pooled as repeated measurements of the same experiment. The earlier MusicEEG
 174 LaBraM uniform-grid increment is 0.0182 [-0.0161, 0.0452], an uncertain auditory case. We retain
 175 it rather than use Urban to erase a non-replicating result. Validation-frozen resource-conditioned
 176 model selection also lacks a stable mean CCC benefit (Appendix C.1); conditional ranking summaries
 177 are not evidence of a proven deployment-selection advantage.

178 Identity probes and subspace perturbations remain representation diagnostics, not causal evidence.
 179 The ds006866 extension concerns conditions, not unknown image identities. DENS is missingness-
 180 limited; SEED-V is an authorized supplement, not an additional openly redistributed benchmark.
 181 EMOD’s repaired single-seed learning check is distinct from its frozen audit and from complete
 182 original-task replication; mdJPT has no completed numerical audit here. These boundaries re-
 183 strict model and domain coverage without making positive EEG performance a prerequisite for an
 184 evaluation contribution.

185 5 What the Tool Can Reuse and Check

186 We execute functions at pinned EEGain and LibEER commits, using eligible EmoEEG-MC trial
 187 identities and three generated marker windows per trial. For LibEER we also execute the precise
 188 empty-validation branch in its EEGNet training entry point; we do not train the external classifier
 189 [13, 18]. Table 4 describes two scope-sensitive examples. An ordinary split checker detects these
 190 observed split violations too. The resource checker’s additional role is checking declared predictor
 191 inputs and execution provenance, not claiming these violations as a novel detector’s exclusive
 192 discovery.

Table 4: Executed external paths and reusable checks. The audit concerns the specified code configuration and claim, not the validity of an original paper’s reported model scores.

Record	Observable finding	Consequence for the requested claim
LibEER empty-validation path	The exercised branch assigns test arrays to validation.	Blocks validation-based selection claimed to exclude test trials.
EEGain window validation split	Generated windows of a real trial can enter both training and validation.	Blocks an additional validation-trial holdout; the same trace still passes test-participant holdout.
Matched-resource toy	Only the EEG predictor receives personal calibration labels.	Blocks a same-label-resource comparison despite clean trial splits.
Missing-provenance toy	Preprocessing fit-trial evidence is absent.	Reports unverifiable, rather than treating absence of evidence as a pass.

193 Twenty native or scope-qualified cases contain ten requested-rule violations and ten clean or correctly
 194 qualified cases. Both checkers give the corresponding decisions. Separately, 70 developer-designed
 195 resource fixtures contain 50 injected violations, ten clean cases, and ten missing-evidence cases. The
 196 resource checker blocks, allows, and preserves uncertainty respectively. These counts verify finite
 197 rule coverage, not a literature-wide detection rate or independent user study. We do not attribute
 198 injected faults to either toolkit, or infer which configuration produced any published result.

199 **Reusable interface.** The source exposes the following no-EEG workflow audit:

```
200 python -m openaffect_eeg.workflow_audit \
201     --trials trials.tsv --execution execution.json \
202     --contract contract.json --output report.json
```

203 The report names failed checks and counts set differences. Reproduction instructions include a
 204 synthetic example with matched, unmatched, missing provenance, and falsely claimed unseen-user
 205 cases. Separate raw-data commands run model experiments on user-provided licensed data. File
 206 hashes establish byte lineage, not the truth of a supplied UID or upstream training history. The release
 207 also contains a machine-readable rule catalog and a source-linked claim-repair record: across five
 208 EEGain folds, the same pinned split function and trial universe block a joint subject-plus-validation-
 209 trial claim, while the narrower held-out-test-participant claim is allowed. This repairs the scope of
 210 the claim; it does not repair or invalidate an external model score.

211 6 Discussion and Limitations

212 **The judgement corrected.** We correct an *attribution rule*: an EEG system’s advantage over an
 213 uncalibrated prior is not, without resource parity, its incremental EEG value. The experiments show
 214 this distinction is numerically material in the evaluated tasks. They do not establish that earlier
 215 publications actually used that unmatched comparison. Nor does every existing subject-independent
 216 protocol become invalid: a properly scoped zero-calibration claim remains meaningful. The proposed
 217 resource ledger refines that claim rather than dismissing it.

218 **When is the method useful?** A researcher comparing encoders can hold test and labelled resources
 219 fixed; a system designer can report a user-calibration budget alongside performance; and a reproducer
 220 can distinguish a valid trial split from a resource-matched comparison. These are supported workflow
 221 capabilities. We have not established broad adoption, independently measured usability, improved
 222 model selection, or an economic benefit from EEG acquisition. A paired no-EEG comparator is a
 223 baseline for the stated task, not a clinically validated alternative device.

224 **Limits of the empirical claim.** The latest analysis is retrospective, covers selected identity pairs,
 225 and uses bounded training settings. Frozen representations and post-fit calibration do not span all
 226 fine-tuning or adaptation methods. Learned preprocessing excludes test trials, but stimulus-dose
 227 injection can change validation-side identity relations; we do not claim every validation identity
 228 stays isolated. The participant and stimulus dose axes are resource interventions on a trial graph,
 229 not randomized interventions on latent affect. Extra normative labels and unverifiable pretraining

230 exposure must remain visible in the contract. Uniform-grid averages can obscure heterogeneous
 231 cells, hence the complete release; intervals condition on fixed fits and have small-cluster limitations.
 232 The independently collected tasks differ in elicitation and target definition, so they do not establish
 233 universal EEG-emotion behaviour.

234 7 Artifacts, Ethics, and Conclusion

235 Code, aggregate results, split rules, source versions, model/data cards, synthetic examples, and
 236 evidence hashes accompany the work. An anonymity-scanned, hash-manifested review archive is
 237 built with the manuscript; its exact archive must be placed at the accessible code URL supplied to
 238 reviewers. Developer-run clean-environment acceptance is reported separately from independent-
 239 human acceptance, which is not claimed. EEG recordings, derived individual features, trial-level
 240 private predictions, stimulus materials, and model weights are not redistributed. Original consent
 241 and licensing apply; Urban’s snapshot contains inconsistent license wording, documented without
 242 redistributing its recordings. Identity probes must not be used to identify participants, and self-report
 243 prediction is not access to a clinical state or private emotion.

244 OpenAffect-EEG provides a fixed-support, resource-matched comparison and an executable check of
 245 its observable assumptions. Its main finding concerns what a score contrast warrants: total system
 246 improvement and EEG added value must be reported separately. Uncertain matched increments are
 247 retained as uncertain; the contribution is improved evaluation specificity and reproducibility, not a
 248 requirement that EEG either succeed or fail.

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A Data Sources and Trial Reconstruction

A.1 Version-pinned sources

OpenAffect-EEG uses fixed OpenNeuro snapshots rather than a floating latest version: MusicEEG ds002721 version 1.0.3, DENS ds003751 version 1.0.6, EmoEEG-MC ds005540 version 1.0.7, and external validation ds006866 version 1.0.0 [8, 23, 25, 30]. Snapshot manifests, byte counts or annex keys, and derived trial-table hashes record the source state. Across the 4.0 datasets we reconstruct 39642.0 trials from 279.0 dataset-qualified participants; 38369.0 trials have complete harmonizable continuous targets. The core tier contains 131.0 participants, 4135.0 trials, 369.0 qualified stimulus identities, and 2865.0 supervised trials.

Table 5: Source and redistribution boundaries. License entries describe the fixed source metadata; companion media and code retain separate terms.

Dataset	OpenNeuro snapshot	Companion boundary	Benchmark release
MusicEEG	CC0	OSF metadata/audio: CC BY 4.0	Metadata, splits, scores
DENS	CC0 field; ODbL/DbCL conflict	Videos not mirrored	Conservative attribution
EmoEEG-MC	CC0	Stimuli/code/checkpoints separate	No EEG, DE, media, weights
ds006866	CC0	OSF table: CC BY 4.0; source images separate	No EEG or images

Table 6: Observed-data and probe-coverage disclosure. EEG feature counts are applied after a frozen trial assignment; missing signals never redefine a split. Probe coverage is the test fraction whose class was observed in fit.

Dataset	Labeled	EEG features	Main role	Stimulus probe
MusicEEG	1,240	1,240	Transparent spectral audit	0.8708
DENS	365	162	Transparent, missingness-limited	0.9826
EmoEEG-MC	1,260	1,260	DE and foundation case study	1.0000
ds006866	35,504	35,504	Condition-level external audit	N/A

The DENS gap originates from unreadable public signal windows rather than an explicit target filter, but the resulting complete-case set is not ignorable. Only 162.0 of 365.0 label-eligible trials are available (missing fraction 0.5562); 7.0 participants have no available trial, and the available-versus-missing valence standardized mean difference is 0.1895. Missingness summaries and probe seen-class fractions are published per dataset and seed. Subject probes in the LaBraM case study have complete test coverage; permutation labels are shuffled only in the training partition while the assignment, features, estimator, and test labels remain fixed.

MusicEEG. Participants listened to film-music excerpts and completed eight self-report items [7]. The benchmark retains 1,240 task trials. Public events do not directly provide canonical audio filenames, so `stimulus_uid` is aligned from event codes to the public OSF Set 1 after checking uniqueness, code range, and the availability of audio and normative ratings for every candidate. Because the README, event description, and early paper do not fully agree on the material set, this mapping is labeled `audited_inference`, not `source_verified`. The OpenNeuro snapshot declares CC0 and the companion OSF resource declares CC BY 4.0; audio is not redistributed.

DENS. DENS uses naturalistic videos and collects dimensional and categorical ratings [2]. Events and behavior are joined by normalized stimulus name, never row position. Several public `.fdt` lengths disagree with their `.set` headers; unreadable full windows are recorded through an availability mask rather than padded, wrapped, or treated as valid EEG. Public events contain twenty local stimulus names, but the snapshot lacks the documented stimulus and code directories, and the cited AFDI entry cannot be reliably linked to those local identities. They are therefore valid grouping keys but not claims of recovered source films. Conflicting CC0 and ODbL/DbCL statements are handled under the more conservative attribution and share-alike boundary.

EmoEEG-MC. The dataset combines video viewing and text-guided imagery [31]. A published behavioral video_name field repeats one erroneous value for some participants and cannot be joined by row order. The supervised subset is restricted to globally verifiable stimulus codes, and context direction is cross-checked against source EDF trigger types. Other reconstructed trials remain in provenance with unavailable targets. Public DE arrays, reordered EEG, model features, and checkpoints are excluded from the anonymous package.

External emotion regulation. ds006866 studies social and nonsocial images under neutral viewing, negative viewing, and reappraisal conditions [24]. It contributes 148.0 participants and 35507.0 image-presentation trials, of which 35504.0 have complete valence and arousal. Public events recover

condition but not trial-level image identity, and EEGLAB annotations do not restore it. Stimulus and joint holdout are consequently blocked by source metadata. The OpenNeuro snapshot declares CC0; companion metadata declare CC BY 4.0, while source images can have separate terms. Neither images nor a substitute media index are distributed.

B Target Harmonization

MusicEEG pleasantness and energetic-arousal reports define valence and arousal; the other datasets use their released valence and arousal fields. MusicEEG, DENS, and ds006866 map an original one-to-nine scale by $(x - 1)/8$; EmoEEG-MC maps zero-to-seven by $x/7$. Original values, scale labels, and transformation expressions remain in trial metadata or result provenance.

The supervised target is the participant’s trial-level report. Nominal stimulus category, normative rating, and experimental condition can be inputs, stratification variables, or audit labels, but do not replace the participant target. Trials missing either continuous dimension remain in metadata and are excluded only from metrics requiring both dimensions.

C Split Protocols

Single-factor protocols and the optimistic control. Complete trials are assigned before windowing. Random-trial, subject, and stimulus holdout use 70/10/20 train/validation/test proportions within each dataset, with at least one identity reserved for validation and test in small groups. Random-trial splitting isolates only `trial_uid` and records subject and stimulus overlaps. Strict single-factor splits require an empty train–test intersection for the held identity.

Joint holdout. Subjects and stimuli are independently assigned 60/20/20. A trial is retained only when both identities receive the same partition; cross-partition edges are marked `excluded`. This sacrifices coverage in sparse subject–stimulus graphs to obtain a genuine double cold start. The ratio was fixed before model evaluation because smaller validation identity sets did not provide adequate validation coverage across seeds.

Support-matched identity-overlap control. For each joint-holdout seed, the control freezes the exact test trial IDs and matches the joint split’s train and validation row counts. It then samples those training-side rows so that both test subjects and test stimuli occur in the fit partition. Consequently, the control preserves test support and sample counts while changing training-side identity overlap and the associated graph. It is an estimand for the practical effect of available identity structure, not a claim that all higher-order graph properties are identical. Across seeds, test sets contain 32.0–56.0 trials from 4.0–7.0 participants and exactly 8.0 held-out stimuli.

Context and dataset holdout. Directional EmoEEG-MC transfer first holds out subjects. Only source-context trials enter training and validation, and only target-context trials from unseen subjects enter test. Video-to-imagery and imagery-to-video are evaluated separately. Context and dataset holdout treat a complete domain as a group.

All main representation audits use 5.0 fixed seeds. Each assignment stores trial roles and an audit JSON. Builders check UID uniqueness, nonempty identity fields, legal roles, and the required empty identity intersections. Models apply target and feature-availability masks only after loading an assignment, so representation missingness cannot silently alter the split.

C.1 Held-out deployment-selection utility

To test a practical consequence rather than assume that ranking changes help, we compare a validation-cell selector with one global validation-mean selector. Both receive identical predictions from seven candidates: a personalized EEG-free prior and direct/combined predictions from spectral features and frozen pretrained/random LaBraM. Candidates, dose grids, role seed and rules are fixed before this analysis. No candidate is removed based on test performance.

Participant and stimulus identities have disjoint fitting, inner tuning, selector-validation and test roles. All trials containing the other phase’s participant or stimulus identities are quarantined, including

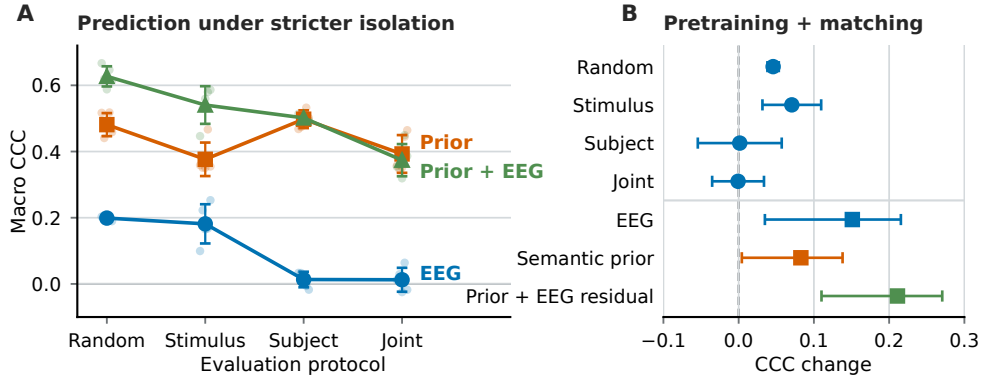


Figure S1: Protocol-level audit retained as supporting evidence. **A**: five fixed split seeds and their mean macro-CCC for semantic prior, direct EEG, and prior-plus-EEG residual; random-trial splitting is an optimistic control. **B**: pretrained versus fixed-random LaBraM comparisons and support-matched identity-overlap effects. These estimates motivate the response-surface analysis but do not by themselves separate participant calibration from repeated-stimulus exposure.

Table 7: Held-out deployment-selection comparison. Both policies use the same validation predictions. Differences are conditioned minus global, uniformly averaged across resource cells; intervals are paired crossed 95% intervals conditional on fits and choices. CCC is primary; MAE is secondary and lower is better.

Dataset	Metric	Global	Conditioned	Difference [interval]
EmoEEG-MC	CCC	0.34221	0.34477	+0.00256 [-0.00113, 0.00601]
EmoEEG-MC	MAE	0.19523	0.19464	-0.00059 [-0.00103, -0.00013]
MusicEEG	CCC	0.24712	0.24600	-0.00113 [-0.01101, 0.01027]
MusicEEG	MAE	0.29405	0.29021	-0.00384 [-0.00675, -0.00094]

resource trials. Assignments use constant placeholders for target-block labels. Ridge tuning uses only inner tuning outcomes; selector choices are hashed before test evaluation. Five resource seeds share the same role partition and target support, not five independent cohorts. The EEG-free and EEG predictors receive identical calibration labels and stimulus-label resources.

EmoEEG-MC has 48 trials from six participants and eight stimuli in each of the two phases. MusicEEG has 25 validation trials (six participants, eight stimuli) and 23 test trials (six participants, nine stimuli), after maximum-dose feasibility filtering. The sparse support limits precision and coverage. All 2,000 paired crossed draws are valid. They condition on fits and validation choices, not selector-training uncertainty. Source, assignment and prediction hashes are verified, and a separate centered-moment implementation reproduces the scores and intervals. This is developer verification, not an independent human reproduction.

Neither primary CCC interval excludes zero. Both secondary MAE differences are small and negative, but no practical-importance threshold was established; they cannot replace the primary endpoint or identify EEG added value. The global choices are random-LaBraM combined prediction for EmoEEG-MC and spectral combined prediction for MusicEEG. These choices describe this candidate set and validation block, not population-level model superiority. This analysis uses datasets already examined during project development: role isolation protects the current selection procedure, but is not untouched external confirmation.

C.2 Identity-exposure response surface

Identity availability is not represented as one binary variable. A deployment state can separately specify whether the participant ID is known, whether unlabelled target-participant data are available, how many labelled calibration trials are available, whether calibration comes from the same session,

Table 8: Maximum-resource predictions with resource-matched controls. P: prior; PC: calibrated prior without EEG; E: prior plus population EEG residual; EC: calibrated E. Brackets are paired crossed-identity 95% pointwise intervals. These are not equivalence tests.

Representation	P	PC	E	EC	EC – PC [95% CI]
<i>EmoEEG-MC</i> ($p=8, s=8$)					
DE	0.4809	0.5863	0.4416	0.5599	-0.0265 [-0.0640, 0.0178]
EEGNet	0.4809	0.5863	0.4863	0.5955	0.0091 [-0.0059, 0.0307]
LaBraM pretrained	0.4809	0.5863	0.4676	0.5748	-0.0115 [-0.0496, 0.0298]
LaBraM random	0.4809	0.5863	0.4629	0.5711	-0.0152 [-0.0553, 0.0179]
CBraMod pretrained	0.4809	0.5863	0.4590	0.5686	-0.0177 [-0.0472, 0.0235]
CBraMod random	0.4809	0.5863	0.4817	0.5840	-0.0023 [-0.0327, 0.0262]
EMOD frozen	0.4809	0.5863	0.4738	0.5819	-0.0045 [-0.0359, 0.0258]
EMOD random	0.4809	0.5863	0.4530	0.5687	-0.0176 [-0.0669, 0.0278]
<i>MusicEEG</i> ($p=8, s=4$)					
Band power	0.1740	0.2930	0.1799	0.2988	0.0058 [-0.0131, 0.0253]
LaBraM pretrained	0.1740	0.2930	0.2085	0.3070	0.0141 [-0.0332, 0.0569]
LaBraM random	0.1740	0.2930	0.1805	0.2941	0.0011 [-0.0531, 0.0587]

and whether the test stimulus, dataset, and device occurred during fitting. The present response surface varies exactly two labelled resources: p , complete calibration trials per target participant, and s , other-participant ratings per test stimulus (or experimental condition where stimulus identity is unavailable). Released normative stimulus metadata, where present, is held fixed and reported as a separate prior rather than counted as s . It fixes unlabelled target-participant adaptation at zero. Session and device are not separately isolated by the available metadata and are therefore blocked claims, not implicit successes.

For each seed, we begin from one strict participant–stimulus joint assignment. The test set is restricted once to participants and stimuli supporting the predeclared maximum doses; its trial IDs are then hash-fixed for every (p, s) cell. Participant calibration candidates contain a test participant under a non-test stimulus. They never enter population-model fitting. Repeated-stimulus candidates contain a test stimulus rated by a non-test participant. Dose samples are deterministic and nested, so every lower dose is a subset of every higher dose on the same axis.

Adding s repeated-stimulus rows would otherwise increase sample size. Each added row therefore replaces one base-training donor selected deterministically within context and nearest in the two training targets. Population-training size is exactly constant across cells, while validation and test trials are unchanged. Donor selection reads training labels only. The intervention intentionally changes which stimulus identities are represented during fitting; it does not claim all graph or target-distribution properties are causally matched.

Let f_s be a population predictor fit with stimulus exposure dose s . For a target participant i , labelled calibration produces only the post-fit offset

$$\hat{\delta}_{i,p} = \frac{1}{p} \sum_{j \in \mathcal{C}_{i,p}} (\mathbf{y}_{ij} - f_s(\mathbf{x}_{ij})), \quad (5)$$

with zero offset at $p = 0$. Test prediction is $f_s(\mathbf{x}_{i,t}) + \hat{\delta}_{i,p}$. Thus a nonzero p cell is explicitly a known-participant, labelled-calibration result; it is not subject independent. The same construction is applied to the prior-plus-EEG predictor and to the EEG-free prior with exactly the same calibration rows. Population and personalized outputs are stored separately.

Matched model extension and reproducibility. The EmoEEG-MC extension evaluates EEGNet and frozen EMOD on every existing assignment, asserting identical test IDs, targets, priors, and calibrated priors. EEGNet selects its epoch count by validation MAE, then refits on train plus validation; target-participant calibration does not update network weights. Direct and residual models are fitted separately. Population fits are reused across participant doses only after verifying unchanged fit assignments. The compact model uses the previously specified custom EEGNet-style architecture with zero adversarial loss. All preprocessing statistics are estimated from its fit data.

EMOD uses official repository revision 3a219f51ba4f90b9e2d22824cfb90dde77a9875c. Its checkpoint has legacy pooling/projection names; the adapter records a one-to-one name mapping and then requires a strict match of all tensors. It uses the benchmark’s 100-Hz raw archive, three nonoverlapping ten-second crops, and mean pooled official projection embeddings. Channels absent from the official channel vocabulary are excluded and listed in the extraction manifest. Input scaling/preprocessing and head adaptation differ from the paper’s full fine-tuning pipeline. This audits the supplied frozen weights, not a claim that EMOD’s original reported SOTA performance was reproduced. Neither third-party weights nor raw feature arrays are redistributed.

Failed native-task learning check. A separate bounded SEED-V check used the official SEEDV model branch and checkpoint backbone with 62 channels, 200 Hz, microvolt inputs and one-second windows. Within each participant/session, the first five, middle five and last five trials formed train, validation and test. Across 48 source files there were 117,744 windows; 40,352 test windows represented 240 trials from 16 participants. A single fixed seed, 100 epochs, and validation-kappa checkpoint selection yielded test balanced accuracy 0.20 and kappa 0: all test windows received one class. Prepared-input hashes, test identities, finite gradients and recomputed metrics were verified, but native-task learning was not recovered. The released SEEDV branch uses a large multilayer head, whereas the paper describes a linear classifier; the per-task selected fine-tuning recipe is not fully specified. This failed compatibility check neither reproduces nor refutes the published scores. It cannot validate the frozen regression audit as a full SOTA test. CUDA pooling backward is nondeterministic despite fixed seeds; this is recorded with the aggregate training history.

Training-only diagnosis and repair. We retained that failed run and reproduced its numerical failure on 50 fixed training windows, ten per class, without validation/test-based parameter selection. With classifier learning rate 10^{-3} , the first update increased the first dense layer’s output standard deviation from 0.773 to 32.280; by step 20, the second hidden ELU produced constant outputs. The official SEEDV head flattens 39,680 features into a 2,480-unit layer. Keeping the architecture, inputs, initialization, optimizer and backbone learning rate 5×10^{-4} unchanged, reducing only the classifier learning rate to 10^{-5} prevented saturation and fitted the diagnostic batch by step 50. This establishes a configuration-specific numerical remedy, not an emotion-mechanism finding. All 48 training-label files matched the official constants. Reprocessing 100 training windows across five source files, including the format-fallback and largest-amplitude cases, matched the prepared float32 inputs exactly; this is selected-file parity, not exhaustive independent data validation.

The repaired run completed the fixed 100-epoch budget. Validation kappa selected epoch 21, after which the test set was evaluated once. Recomputed test BACC was 38.62%, kappa 0.2348 and weighted F1 39.43%; all five classes were predicted. The published five-run BACC is $41.24 \pm 0.86\%$ [6], leaving a 2.62 percentage-point gap to that mean. This single-seed check demonstrates recovered learning, not statistical equivalence or a five-run reproduction. It uses within-participant classification, not the self-report exposure response surface, and cannot establish unseen-user transfer or pretrained benefit without a matched random-initialization comparison. Aggregate traces, input/source hashes and a fail-fast input-sensitivity check accompany the repair. The original failed results remain archived.

Standard implementation and stronger calibration control. The added standard regression model uses Braindecode 1.2.0 EEGNet with $F_1 = 8$, $D = 2$, $F_2 = 16$, temporal kernels 64/16, dropout 0.25, and two raw regression outputs. AdamW uses learning rate 0.001 and weight decay 0.0001; validation MAE selects among at most 100 epochs with patience 12. All 25 population states and 125 dose cells use the archived assignments. Its 50 direct/residual fits have finite logged objectives and gradients; selected epochs range from 1 to 24, with 14 fits selecting epoch one. Synthetic learning and training-only tiny-subset fitting checks pass, but do not demonstrate optimal benchmark tuning or held-out emotion information.

An additional empirical calibration control estimates within-participant residual variance σ^2 and nonnegative noise-corrected between-participant mean variance τ^2 from population-fit data. It multiplies every calibrated offset by $w_p = p\tau^2/(p\tau^2 + \sigma^2)$, with $w_0 = 0$. These are heuristic prediction weights, not identified latent variance components. The same weights apply to *PC* and *EC*. Table 9 reports uniform cell means; the historical dose-weighted integrals remain separate secondary summaries. Neither calibration rule establishes a positive surface-average EEG increment for the displayed models at the reported precision.

Table 9: Implementation and calibration sensitivity. Entries are uniform-grid mean matched EEG added value, $CCC(EC) - CCC(PC)$, with 95% paired crossed intervals conditional on fitted predictions. Shrinkage weights are estimated only from population-fit residuals and shared by EEG and non-EEG offsets. These are not maximum-dose scores or equivalence tests.

Dataset	Representation	Mean offset	Shrunk offset
EmoEEG-MC	Compact EEGNet-style	0.0089 [-0.0034, 0.0254]	0.0067 [-0.0064, 0.0253]
EmoEEG-MC	Standard EEGNet	0.0006 [-0.0036, 0.0054]	0.0006 [-0.0044, 0.0060]
EmoEEG-MC	LaBraM	-0.0100 [-0.0392, 0.0279]	-0.0098 [-0.0419, 0.0308]
EmoEEG-MC	Frozen EMOD	-0.0001 [-0.0280, 0.0247]	-0.0005 [-0.0302, 0.0263]
MusicEEG	Band power	0.0042 [-0.0094, 0.0188]	0.0052 [-0.0269, 0.0335]
MusicEEG	LaBraM	0.0182 [-0.0161, 0.0452]	0.0247 [-0.0176, 0.0568]

Table 10: Companion error contrasts for the same matched resources. Entries are uniform-grid mean $MAE(EC) - MAE(PC)$ with paired crossed 95% intervals. Negative values favor the EEG branch; positive values mean larger error. These marginal intervals do not control error across all representations and calibration variants.

Dataset	Representation	Mean offset	Shrunk offset
EmoEEG-MC	Compact EEGNet-style	-0.0008 [-0.0030, 0.0011]	-0.0002 [-0.0032, 0.0023]
EmoEEG-MC	Standard EEGNet	0.0004 [-0.0004, 0.0012]	0.0005 [-0.0005, 0.0015]
EmoEEG-MC	LaBraM	0.0059 [0.0001, 0.0114]	0.0056 [-0.0009, 0.0122]
EmoEEG-MC	Frozen EMOD	0.0032 [-0.0025, 0.0100]	0.0029 [-0.0029, 0.0099]
MusicEEG	Band power	0.0013 [-0.0021, 0.0044]	0.0033 [-0.0045, 0.0108]
MusicEEG	LaBraM	-0.0006 [-0.0071, 0.0067]	-0.0010 [-0.0100, 0.0096]

Revised uncertainty. The paired tables use 2,000 shared crossed-identity draws across fixed splits. The same draws compare every predictor, dose, and model; a degenerate draw is not repaired by silently dropping a seed. The tables expose valid draw counts. The legacy resource-effect table below preserves the original seedwise bootstrap for traceability and is not the revised estimator used in the main text. In particular, MusicEEG’s revised joint interval crosses zero. For ds006866, each fixed split holds out one condition; its condition factor cannot support a within-split resampling claim about new conditions.

MusicEEG’s frozen grid is five participant doses by four stimulus doses, ending at $s = 4$. Raising s to eight cannot preserve the same test support in four of the five seeds: their available other-participant observations are insufficient. Only one seed supports that extension without dropping test trials. We do not fill unavailable cells or compare a changed test set as if it were the same response surface.

The artifact emits one deterministic claim card per cell. It records ID and label requirements and blocks latent-emotion disentanglement, causal identity removal, unseen-participant claims when $p > 0$, unseen-stimulus claims when $s > 0$, and session/dataset/device transfer unless those factors are separately held out. This is rule-based compilation from split lineage, not free-form language-model judgment.

D EEG Representations and Models

Transparent spectra. Core transparent features use delta, theta, alpha, beta, and gamma DE or Welch log power. Global views aggregate channels; channel views retain the channel-by-band layout. Standardization is fitted on training data only. Ridge regularization is selected by validation MAE and refitted on train plus validation. Single-band, leave-one-band, single-region, and leave-one-region views test predictive sensitivity, not source localization or causal anatomy.

CBraMod. Official pretrained and same-architecture random encoders receive identical EmoEEG-MC trials. Each 30-second, 64-channel, 200-Hz trial is organized into one-second patches; channel and time tokens are mean-pooled into a 200-dimensional trial representation. Source commit, checkpoint SHA-256, trial UIDs, and feature hashes accompany the artifact. The encoder is frozen and only the common downstream head is trained.

Table 11: Legacy fixed-support identity-exposure summary (original seedwise bootstrap, retained for traceability). Values are macro-CCC changes from the zero-resource cell; brackets are crossed participant–stimulus 95% intervals. For ds006866, S denotes experimental-condition exposure because source metadata do not expose trial-level image identity.

Dataset	Representation	Zero	Maximum	Surface AUC	ΔP	ΔS	$\Delta(P+S)$
EmoEEG-MC	LaBraM pretrained	-0.0052	0.5748	0.4604	0.1788 [0.0712, 0.2307]	0.4728 [0.2900, 0.5420]	0.5800 [0.4072, 0.6219]
MusicEEG	LaBraM pretrained	0.1026	0.3070	0.2302	0.0953 [-0.0097, 0.2037]	0.1060 [-0.0950, 0.2798]	0.2045 [0.0157, 0.3320]
ds006866	Band power	–	–	–	0.2856 [0.2132, 0.3461]	-0.0014 [-0.0071, 0.0042]	0.3256 [0.2694, 0.3685]

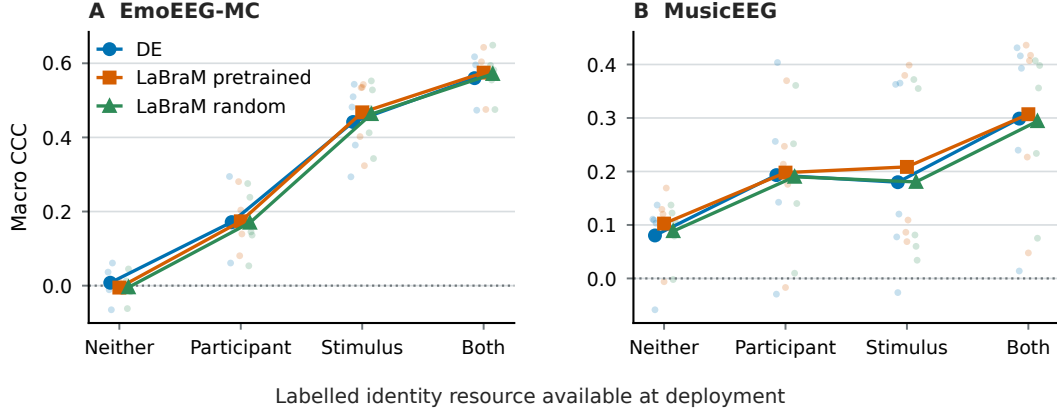


Figure S2: Deployment corners for the three main representations. Large markers and connecting lines show seed means; faint points are the five fixed split seeds. “Participant” uses the maximum labelled calibration dose with unseen stimuli, “Stimulus” uses the maximum repeated-stimulus dose with unseen participants, and “Both” uses both resources. The large MusicEEG spread is why endpoint means are not treated as independent replications.

588 **LaBraM.** LaBraM uses the official channel-location map and base checkpoint. Thirty one-second
589 patches are processed as three eight-patch windows and one six-patch window, then pooled with
590 patch-count weighting into a 200-dimensional trial representation. Pretraining masks and decoder
591 heads are excluded, while the official final normalization path is retained. The random control
592 instantiates the same architecture without loading a checkpoint.

593 **Historical compact EEGNet-style control.** The custom task-specific control is trained from
594 EmoEEG-MC waveforms resampled with anti-aliasing to 100 Hz. Each epoch uses one deterministic
595 10-second training crop; validation and test average three nonoverlapping crops. The network
596 uses temporal convolution, depthwise spatial convolution, separable temporal convolution, and an
597 eight-dimensional latent. It tests architecture dependence, not an exhaustive architecture search.

598 E Predictors, Debiasing, and Identity Probes

599 **Predictors.** `direct_eeg` predicts normalized targets from EEG. `semantic_stimulus_prior`
600 combines one-hot nominal category, context, and source stimulus-intensity code with lower-
601 cased TF-IDF description unigrams/bigrams (at most 1,024 features, sublinear term frequency).
602 A Ridge model is fitted to one mean target row per training stimulus; alpha is selected from
603 $\{0.01, 0.1, 1, 10, 100, 1000\}$ by trial-level validation MAE. The intensity code identifies source
604 stimulus variants and never uses a held-out participant’s rating. The historical registry identifier
605 `disentangled_prior_plus_eeg_residual` uses a training-only stimulus report mean for seen
606 stimuli and the semantic estimate for unseen stimuli, forms leave-one-trial-out residual targets, pre-
607 dicts them from EEG, and adds the hybrid prior. All branches share the assignment and validation
608 rule; none reads test targets. The identifier denotes arithmetic residualization and does not claim
609 causal or latent-state disentanglement.

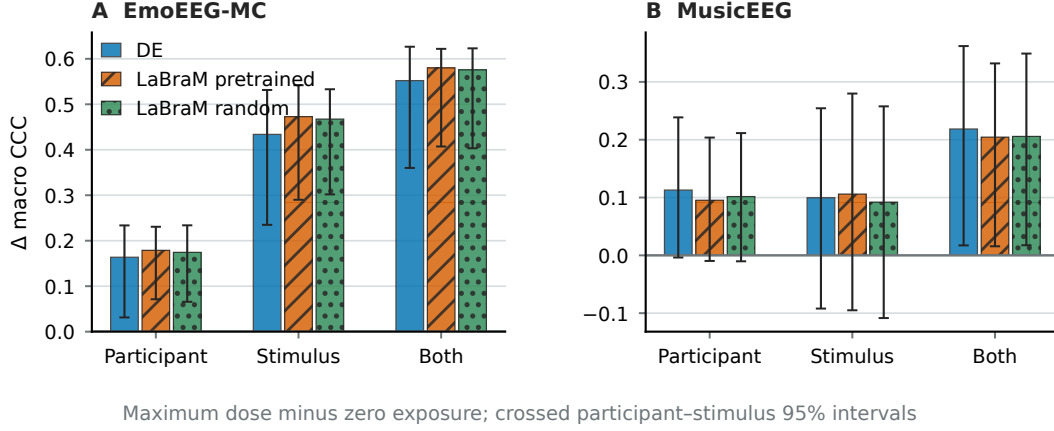


Figure S3: Maximum-dose minus zero-resource changes. Bars distinguish transparent features, pretrained LaBraM, and fixed-random LaBraM; whiskers are crossed participant–stimulus bootstrap 95% intervals. EmoEEG-MC supports positive marginal and joint changes under the legacy estimator. The revised shared-identity interval for MusicEEG’s joint change crosses zero; this legacy plot is not evidence of a confirmed external resource effect.

Table 12: Dataset–protocol applicability matrix. “Blocked” denotes missing source identity metadata, not model failure.

Dataset	Random	Subject	Stimulus	Joint	Context	Dataset
MusicEEG	Yes	Yes	Yes	Yes	N/A	Yes
DENS	Yes	Yes	Yes	Yes	N/A	Yes
EmoEEG-MC	Yes	Yes	Yes	Yes	Yes	Yes
ds006866	Yes	Yes	Blocked	Blocked	Yes	N/A

Debiasing counterfactuals. We evaluate stimulus centering, conditional covariance penalties, response-preserving stimulus-subspace projection, gradient-reversal residual networks, and adversarial EEGNet. Every method contains a zero-strength setting; adversarial strength, projection rank, and epoch are selected by validation residual MAE only. Test-time residual prediction does not use stimulus ID. A successful counterfactual must jointly improve target performance and reduce the relevant identity effect. Variance shrinkage or inconsistent seed direction is reported as a negative result.

Identity probes. Dataset identity is predicted under subject holdout; stimulus identity under subject holdout is evaluated only on classes observed in training; subject identity under stimulus holdout is likewise evaluated on training-seen people. Each probe uses a matched training-label permutation and reports balanced accuracy, class count, uniform chance, coverage, and per-seed values.

Subject-subspace intervention. For each stimulus-holdout seed, between-subject centroid differences are estimated on training data, decomposed by SVD, and the leading directions are removed from fit and test LaBraM representations. The requested removal grid is $\{0, 1, 2, 4, 8, 16, 24, 29\}$. At every rank, we also remove the same number of training-estimated PCA directions and seeded random orthonormal directions. Semantic, direct, residual, and subject-probe heads are refitted with the same validation-only selection. The zero-removal point is the nested control. This intervention destroys linearly decodable subject structure but can also remove participant-related affect variation; it is not interpreted as a pure nuisance projector.

F Statistical Analysis

Continuous predictions report valence, arousal, macro CCC, MAE, and RMSE. Identity tasks report balanced accuracy and, where available, macro-F1. Model comparisons are paired on identical trial-level predictions. In the legacy non-surface analyses, the hierarchical bootstrap first resamples

Table 13: Executed analysis coverage. This table is intentionally distinct from protocol applicability: “supported” does not imply that every model was run in that cell. V/I denotes directional video–imagery transfer.

Dataset	Representation or analysis	Executed protocols
EmoEEG-MC	DE, CBraMod, LaBraM affect heads	Random, subject, stimulus, joint, V/I; exposure grid
EmoEEG-MC	EEGNet	Joint; complete exposure grid
EmoEEG-MC	EMOD frozen and random	Complete exposure grid
MusicEEG	Band power, LaBraM pretrained/random	Complete feasible exposure grid
MusicEEG, DENS, EmoEEG-MC	Transparent identity probes	Dataset, subject, stimulus as applicable
ds006866	Channel/global bandpower	Subject holdout and condition probe

Table 14: Representation controls and training state.

Representation	Trial view	Pretraining control	Downstream state
DE / Welch	Global, channel, band, region	N/A	Validation-selected Ridge
CBraMod	Frozen 200-D embedding	Same architecture, random	Frozen encoder
LaBraM	Frozen 200-D embedding	Same architecture, random	Frozen encoder
EEGNet	Raw deterministic crops	Zero adversary	End-to-end control
EMOD	Frozen 128-D embedding	Same architecture, random	Validation-selected Ridge

633 evaluation seeds and then complete held-out identity clusters; joint holdout separately resamples
634 crossed subject and stimulus factors. Those historical intervals use 2,000 resamples and a fixed
635 statistical seed. This is not the revised primary response-surface estimator, which holds split seeds
636 fixed and shares crossed identity weights across every model/cell.

637 With 5.0 seeds, the exact two-sided sign-flip test has a minimum nonzero p -value of 0.0625. We
638 therefore interpret bootstrap intervals as effect uncertainty and the sign-flip test as a finite-seed
639 resolution statement. Historical targeted robustness analyses comprise the support-matched CCC
640 contrast, subject-subspace endpoint CCC contrast, unseen-stimulus LaBraM pretraining contrast,
641 and external condition-prior paired contrasts. These were selected during benchmark development
642 and reviewer-driven audit rather than preregistered. Other reported cells are exploratory. Seeds
643 represent Monte Carlo split variability; participant, stimulus, or crossed cluster resampling supplies
644 the corresponding generalization uncertainty. Intervals are pointwise and are not presented as
645 simultaneous family-wise coverage.

646 Under the same support-matched contrast, the semantic stimulus prior changes by 0.0827 CCC and
647 the hybrid stimulus prior by 0.0991. These branches are reported separately so the full-pipeline effect
648 is not attributed to EEG alone. Component ablations further separate nominal category, context,
649 source intensity code, and description text (Table 17).

650 The strong nominal-category effect under unseen-stimulus and joint holdout shows that the semantic
651 prior’s transferable score is largely experiment-label structure. Description text contributes under
652 subject holdout, whereas context, intensity, and description do not have stable independent joint-
653 holdout effects. These are predictive ablations, not causal interventions on stimulus content.

654 G External Validation Details

655 The initial subject-holdout analysis found that the condition prior exceeded EEG-only macro-CCC by
656 0.3457, with interval [0.304, 0.3884]. The v2 audit extends this case to exactly matched bandpower,
657 pretrained-LaBraM, and fixed-random-LaBraM representations under five protocols. Table 19 reports
658 every frozen cell rather than only the favorable repeated-participant setting.

Table 15: Optimization and selection budget. Every choice is made on the validation partition and then reused for the paired test comparison.

Component	Validation search	Training budget
Ridge prior/EEG/residual heads	Six alphas from 10^{-2} to 10^3	Closed-form per seed/protocol
Linear nuisance controls	Five strengths from 0 to 1	Closed-form per strength
Gradient-reversal residual	Six adversary strengths	At most 200 epochs; patience 25
Adversarial EEGNet	Three adversary strengths	At most 100 epochs; patience 12
Subject-subspace audit	Eight ranks, three direction types	Head refit at every point

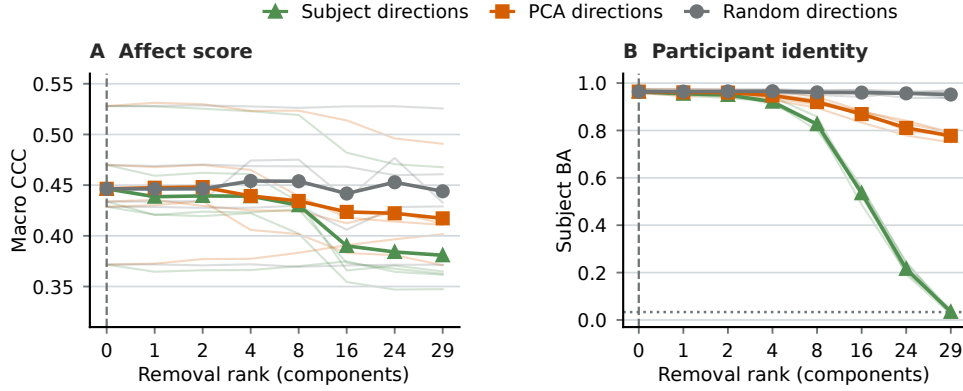


Figure S4: Subject-subspace intervention audit. **A** plots affect macro-CCC and **B** participant balanced accuracy across the fixed removal-rank grid. Faint trajectories are the five evaluation seeds; opaque lines and markers are their means. Triangles remove subject-aligned directions, squares remove equal-rank PCA directions, and circles remove equal-rank random directions. The vertical dashed line is the nested zero-rank control; the horizontal dotted line in **B** is chance. The figure exposes an accuracy–identity trade-off and does not identify participant structure as a pure nuisance factor.

For repeated participants, the hybrid improves over the prior in both directions and all five participant-clustered MAE intervals lie below zero for each representation. Under joint participant–context holdout, no LaBraM hybrid interval establishes an improvement over the prior. Pretrained-minus-random hybrid CCC is -0.0008 and -0.0082 across the two joint directions, with only two of five seeds positive in each. Thus the strict result does not support a pretraining-specific residual advantage.

The near-perfect repeated-participant probes establish information available to a fixed linear readout; they do not show that the affect head causally uses identity. Public events expose condition but not trial-level image identity, so stimulus holdout remains blocked. Header-only conformance confirms the same ordered 64-channel EEG set in all 148 recordings; 146 match the base source pattern and two contain one additional, predeclared EMG channel.

H Authorized SEED-V Protocol-Audit Case Study

SEED-V is a stimulus-induced five-class EEG and eye-movement study with 16 participants, three sessions, and fifteen film clips per session [16]. We report a separately authorized raw-EEG analysis as an independent case study rather than a fifth OpenAffect-EEG source. Its source recordings, feature arrays, media, trial predictions, participant codes, and project paths are not redistributed. The public artifact contains only the source-safe aggregate in `paper/generated/seedv_case_study.json`, its input and code hashes, and this interpretation boundary.

The authorized project compared a raw-temporal baseline against a pre-existing SDSA variant under a strict leave-one-participant-out protocol and a joint participant–stimulus protocol. Table 21 reports all frozen cells rather than selecting the favorable joint cell. The strict SDSA contrast is negative, and both bootstrap intervals include zero. Its frozen decision gate therefore fails; the contrast is evidence of protocol sensitivity, not a model-improvement claim. Separately, the authorized project’s strict-to-

Table 16: Selected paired macro-CCC comparisons. Intervals are hierarchical paired-bootstrap 95% intervals.

Comparison	Protocol	Δ CCC	Interval	Scope
LaBraM pretrained – random	Stimulus	0.0706	[0.0396, 0.1143]	Known subjects, new stimuli
LaBraM pretrained – random	Subject	0.0015	Includes zero	New subjects
LaBraM pretrained – random	Joint	-0.0008	Includes zero	New subjects and stimuli
Identity overlap – joint, direct EEG	Matched sup-port	0.1511	[0.0349, 0.2156]	Same test and fit counts
Identity overlap – joint, full pipeline	Matched sup-port	0.2111	[0.1101, 0.2706]	Includes prior and residual
Max subject removal – zero	Stimulus	-0.0656	[-0.107, -0.0272]	LaBraM intervention
Subject removal – equal-rank PCA	Stimulus	-0.0364	[-0.0624, -0.0118]	Direction specificity
Subject removal – equal-rank random	Stimulus	-0.0631	[-0.1028, -0.0263]	Direction specificity
Condition prior – EEG-only	External subject	0.3457	[0.304, 0.3884]	New subjects, repeated conditions

Table 17: Semantic-prior component audit. Positive values favor the full prior; intervals use the holdout identity required by each protocol.

Protocol	Omitted component	Full – ablated CCC	95% interval
Stimulus	Nominal category	0.3049	[0.2089, 0.3416]
Subject	Description text	0.0544	[0.0384, 0.071]
Joint	Nominal category	0.4011	[0.2557, 0.4591]
Joint	Context	0.0012	[-0.004, 0.0064]
Joint	Intensity code	-0.01	[-0.0296, 0.0055]
Joint	Description text	-0.0129	[-0.0529, 0.0261]

681 PaperStyle ladder rises from 0.2786 to 0.5922, so the large protocol gap cannot be attributed to a
682 model variant alone [16].

683 I Independent MusicEEG Protocol Audits

684 To test whether the EmoEEG-MC representation finding is confined to one visual paradigm, we run
685 two prespecified audits on independent MusicEEG [7]. Each uses 1,240 labelled 12-second music
686 trials from 31 participants, five frozen complete-trial splits, and source-safe aggregate reporting. First,
687 a transparent channel log-bandpower Ridge readout reaches macro CCC 0.1031 under random-trial
688 evaluation, 0.0175 with unseen participants, 0.1130 with unseen music but known participants, and
689 0.0227 under joint participant–music holdout. The paired random-minus-joint difference is 0.0804
690 (seed SD 0.0357). Thus holding out music alone does not substitute for holding out participants in
691 this auditory source.

692 The same bandpower representation identifies training-seen participants under unseen music at
693 balanced accuracy 0.9995 (permuted 0.0431; 31-class uniform chance 0.0323). In contrast, the
694 seen-class music probe under unseen participants is 0.0000 balanced accuracy (permuted 0.0033;
695 78.8–91.3% test coverage across seeds). These probes establish available information under their
696 stated coverage, not causal use by the affect predictor.

697 The self-report diagnostic also constrains interpretation. Leave-one-rating-out same-music consensus
698 CCC is 0.1437 for valence and 0.1634 for arousal (1,123 supported ratings each). Descriptive in-
699 sample additive models assign more R-squared to music identity (0.3579 and 0.3484) than participant
700 identity (0.1170 and 0.1782); they are neither test–retest reliability estimates nor causal variance
701 decompositions.

Table 18: Identity-probe summary. Permuted values use the same representation, split, and estimator with shuffled training labels.

Representation and task	Holdout	Observed	Permuted	Chance
LaBraM pretrained, subject	Stimulus	0.9642	0.0192	0.0333
LaBraM random, subject	Stimulus	0.1733	0.0425	0.0333
DE, dataset	Subject	0.8987	0.3913	0.3333
LaBraM pretrained, stimulus	Subject	0.0234	0.0260	0.0238
External channel, condition	Subject	0.2486	0.1676	0.1667
External global, condition	Subject	0.2134	0.1692	0.1667

Table 19: Complete ds006866 v2 affect audit. Entries are five-seed mean macro-CCC. BP denotes 320-dimensional channel bandpower; Pre. and Rand. denote frozen pretrained and fixed-random LaBraM. Hybrid is the transferable condition-family prior plus the EEG residual.

Protocol	Prior	EEG BP	EEG Pre.	EEG Rand.	Hybrid BP	Hybrid Pre.	Hybrid Rand.
Unseen participant	0.3776	0.0369	0.0335	0.0198	0.3425	0.3742	0.3797
Nonsocial → social, repeated participant	0.3757	0.2319	0.2604	0.2539	0.5144	0.5507	0.5476
Social → nonsocial, repeated participant	0.3780	0.2226	0.2526	0.2455	0.5116	0.5486	0.5462
Nonsocial → social, unseen participant	0.3704	0.0208	0.0323	0.0152	0.3318	0.3718	0.3726
Social → nonsocial, unseen participant	0.3792	0.0172	0.0289	0.0220	0.3472	0.3761	0.3844

Second, we apply the frozen official LaBraM base encoder to exactly the same MusicEEG split family. Each trial uses its 19 recorded 10–20 channels and its observed event window; MNE volts are converted to microvolts, common-average referenced within trial, resampled to 200 Hz, and final tokens pooled. There is no padding, cross-trial window, fine-tuning, or feature release. Pretrained and fixed-random initializations use the same 200-dimensional Ridge readout, manifests, five split seeds, and nested validation selection.

Table 22 is a separate foundation-model audit rather than a new model claim. Pretraining has a positive matched macro-CCC point difference in all four audited protocols, including participant and joint holdout, but the corresponding unseen-music subject probe is also substantially higher for pretrained features. This external source therefore qualifies, rather than reverses, the main case study: pretrained features have higher report-prediction point estimates in this auditory source while retaining participant structure, so neither the positive gain nor a single identity probe supports a general invariant- emotion conclusion.

J Compute and Software Environment

Transparent features, linear baselines, probes, paired statistics, artifact compilation, and tests run in a CPU environment on Linux x86-64 with Python 3.11, NumPy 2.4.6, pandas 3.0.5, SciPy 1.17.1, scikit-learn 1.9.0, and MNE 1.12.1. Foundation feature extraction and EEGNet use one NVIDIA RTX 6000 Ada Generation GPU with 49,140 MiB memory, PyTorch 2.11.0, and CUDA 13.0. CPU-only adversarial residual runs use four threads to avoid competing with GPU jobs.

The typed recomputation graph separates feature extraction, evaluation, statistics, and manuscript compilation. Exact historical wall-clock and energy totals were not captured consistently for every preliminary or failed run; the release therefore reports component environments and DAG boundaries rather than inventing a project-wide total. The final cold-run release log is designed to record per-step wall times and will be published with the executable artifact. This limitation does not affect frozen scores but constrains precise compute-cost replication.

K Release, Licensing, and Reproduction Boundaries

The anonymous package contains code, tests, configurations, evaluation cards, dataset and model cards, Croissant metadata, generated table sources, and claim configuration. It explicitly excludes raw EEG, source feature arrays, foundation checkpoints, movie, music, and image stimuli, office documents, Git history, and machine-local paths.

Table 20: Matched ds006866 v2 identity probes. Each cell is observed/permutated five-seed mean balanced accuracy. Chance reflects the fitted class set.

Protocol	Probe	BP	Pretrained	Random	Chance
Unseen participant	Six-condition	0.2448/0.1669	0.1713/0.1667	0.1675/0.1653	0.1667
Nonsocial → social, repeated participant	Participant	0.9910/0.0066	0.9997/0.0074	0.9963/0.0057	0.0068
Social → nonsocial, repeated participant	Participant	0.9890/0.0063	0.9999/0.0045	0.9973/0.0046	0.0068
Nonsocial → social, unseen participant	Condition family	0.4091/0.3238	0.3399/0.3317	0.3306/0.3333	0.3333
Social → nonsocial, unseen participant	Condition family	0.4206/0.3268	0.3384/0.3336	0.3308/0.3315	0.3333

Table 21: Authorized SEED-V aggregate audit. Values are mean trial macro-F1 across the frozen result rows; Δ is SDSA minus raw temporal. No participant-level or trial-level values are released.

Protocol	Raw temporal	SDSA	Δ	95% interval
Strict participant holdout	0.2418	0.2297	-0.0121	[-0.0439, 0.0178]
Joint participant–stimulus holdout	0.1788	0.1856	0.0068	[-0.0154, 0.0292]

metadata mode validates the benchmark contract without source data. smoke uses deterministic synthetic trials to exercise splitting, baselines, probes, and a small registry. full requires an explicit user-provided data root and verifies frozen input hashes and paper assets. The typed recompute plan/run interface covers trial construction, feature extraction, evaluation, probes, statistics, and paper compilation; -resume skips a step only when all declared outputs exist.

The shortest exact verification sequence from a clean checkout is:

```
conda env create -f environment.yml
conda activate openaffect-eeg
python -m pip install -e ".[smoke]"
python scripts/run_benchmark.py metadata \
  --output-root /tmp/openaffect-metadata
python scripts/run_benchmark.py smoke \
  --output-root /tmp/openaffect-smoke
export D=/path/to/OpenAffect-EEG-data
python scripts/run_benchmark.py full \
  --data-root "$D" --output-root "$D/reproduction"
python scripts/build_neurips_submission.py
```

The first two modes never read EEG. Full mode verifies the frozen registry and rebuilds evidence-linked tables; it is not mislabeled as raw feature extraction or model retraining. The separately documented recompute_benchmark.py plan/run commands expose that typed raw-to-paper DAG and require explicit paths to source data, checkpoints, and foundation-model repositories.

Full-mode verification of frozen outputs is not described as a cold download and multi-day retraining run. The existing isolated-environment verification was performed on the same compute host and is not represented as independent laboratory reproduction.

L Machine-Readable Attachments

The full registry retains per-seed identity results, all paired model comparisons, band and region sensitivity cells, prediction-key audits, source hashes, and unsupported protocol states. These files prevent selective compression of null or unfavorable results into the PDF. Region analyses are predictive sensitivity tests and are not interpreted as neural source localization.

M Position Relative to Closest Evaluation Resources

N Latest Resource-Matched Coverage Analysis

This section and the main resource-matched tables use the frozen final_closure_v9 evidence. Earlier sections preserve the original experiments, figures, and their support. They are not additional

Table 22: Independent MusicEEG frozen LaBraM audit. Values summarize five fixed split seeds; Δ is pretrained minus fixed-random macro-CCC. Subject probes predict training-seen participants under unseen music.

Protocol	Pretrained CCC	Random CCC	Δ CCC
Random trial	0.0923	0.0191	0.0732
Unseen participant	0.0697	-0.0006	0.0703
Unseen music	0.0847	0.0145	0.0703
Unseen participant and music	0.0422	0.0085	0.0337
Unseen-music subject probe (BA)	0.9136	0.4239	—
Permuted subject probe (BA)	0.0313	0.0297	chance 0.0323

Table 23: The closest resources address complementary layers of the problem. The final column states the additional evaluation object introduced here, rather than asserting that prior work is invalid.

Resource	Primary evaluation object	Additional OpenAffect-EEG object
Leakage audit [15]	Trial-segment and test-data leakage on DEAP	Subject, stimulus, joint, context, and dataset identity contracts
mdJPT [33]	Multi-dataset affect pretraining and new-dataset transfer	Deployment-specific identity isolation plus matched random encoders
EMOD [6]	Discrete/continuous label alignment in a common V-A space	Self-report prior/residual separation and stimulus provenance states
EEG-FM-Compass [20]	Cross-task FM comparison under subject transfer and calibration	Affect-specific stimulus and condition audits with predictor-linked interventions
FMScope [19]	Subject-axis diagnosis and erasure across clinical EEG-FMs	Joint subject-stimulus-context contracts and report-level affect estimands
EEG concept audit [29]	Encoded-versus-used physiological concepts under erasure	Experiment-metadata priors separated from trial-level self-reports
Emotion protocol audit [28]	Subject-dependent versus subject-disjoint SEED evaluation	Stimulus, joint, context, and dataset isolation with provenance states
OpenAffect-EEG	Executable evidence and applicability contract	Hash registry, claim ledger, cards, blocked cells, and matched controls

independent replications of this panel, and their historical use of “primary” denotes their own frozen analysis plan rather than replacing the present main summary.

N.1 Source and coverage details

EmoEEG-MC uses snapshot 1.0.7; Urban Appraisal uses snapshot 1.0.0 [30, 32]. Emo has 1,260 eligible trials from 30 participants and 42 stimulus identities. Urban uses 3,471 valence trials from 62 participants and 56 images; other scales are not copied into valence/arousal labels. A source recording with confirmed truncation was excluded before prediction inspection, as documented in the frozen coverage manifest. Urban’s dataset-description license says CC0, while its README says CC BY 4.0. We document both declarations and do not redistribute the source recording or image material.

SHA-256 ordering with design seed 20260908 assigns participant and stimulus identities to five blocks. Block b supplies test pairs; block $b + 1$ modulo five supplies validation pairs; population training initially uses identities from the other blocks. Test pairs lie on diagonal blocks only. Coverage means every eligible participant and stimulus is tested once, not every possible trial. The same test support is retained throughout the dose grid. Stimulus-label injection may use participants from the validation block; only the tested participants are categorically excluded from population fitting at all doses. This is not a claim of fully disjoint validation identities after injection.

N.2 Bounded model settings and preprocessing

Ridge R1 selects alpha from $\{0.1, 1, 10, 100\}$ and R2 from $\{1, 10, 100, 1000\}$. Both band-power and frozen LaBraM use the two grids. The historical Emo feature key `de` selects channel log-bandpower;

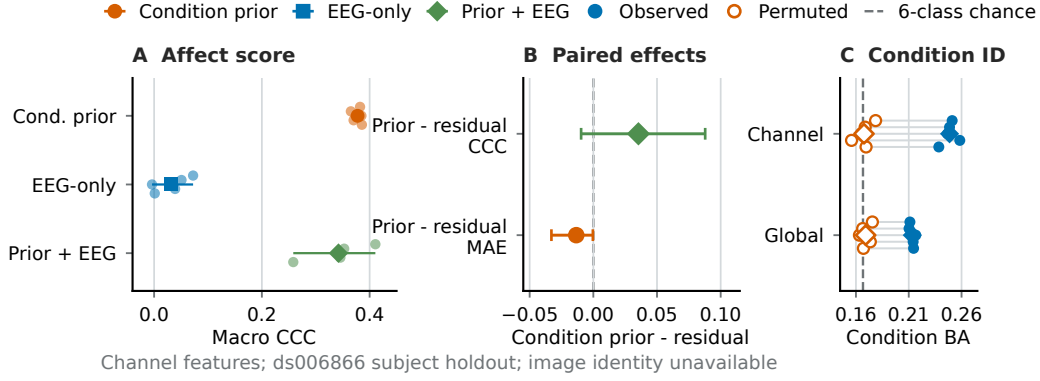


Figure S5: Initial ds006866 subject-holdout condition audit. The condition prior, direct EEG, and condition-plus-EEG residual are reported separately; paired residual effects and condition probes use the five fixed seeds. Image identity is unavailable, so this is condition-level evidence and cannot support a stimulus-holdout claim.

785 it does not make this panel a separate differential-entropy implementation. Standard EEGNet is
786 Braindecode 1.2.0 with F1=8, D=2, F2=16, dropout 0.25, AdamW weight decay 10^{-4} , batch size
787 128, at most 100 epochs and patience 15. L1 and L2 use learning rates 10^{-3} and 10^{-4} . Validation
788 MAE selects training length; the selected length is refitted on train plus validation. Both direct and
789 residual branches are retained. Calibration changes post-fit offsets only; it is not a sweep of few-shot
790 fine-tuning algorithms.

791 Emo uses the existing 100-Hz raw trial archive and ten-second EEGNet crops. Urban uses 64 EEG
792 channels in physical volts, excludes ECG and GSR, applies common-average reference and polyphase
793 resampling from 500 to 100 Hz, and uses the three-second image interval before the rating response.
794 At evaluation, window predictions are averaged within a trial. Downstream learned normalization is
795 fitted without test or calibration trials. Frozen LaBraM uses the project's recorded checkpoint and
796 channel/unit adaptation; its upstream training-data identity provenance is not certified here.

797 N.3 Complete primary and secondary summaries

798 All tables below are generated from the same hashed aggregates as the main tables. No new fitting,
799 bootstrap, or model selection was performed to produce this manuscript revision. The score identity
800 $EC - P = (PC - P) + (EC - PC)$ is checked for all 300 task-setting-dose combinations
801 after block aggregation, with maximum numerical error below 10^{-12} . This validates a numerical
802 identity, not a causal decomposition. Paired intervals are copied from paired contrast outputs, never
constructed by subtracting separate confidence bounds.

Table 24: EmoEEG-MC: uniform-grid EC-PC CCC. Paired pointwise intervals condition on fixed fitted predictions; t-normal is a heuristic sensitivity analysis.

Setting	Mean	Percentile interval	t-normal interval
Band-power R1	-0.0069	[-0.0333, +0.0221]	[-0.0435, +0.0298]
Band-power R2	-0.0091	[-0.0306, +0.0145]	[-0.0385, +0.0202]
LaBraM R1	-0.0320	[-0.0676, +0.0104]	[-0.0842, +0.0201]
LaBraM R2	-0.0137	[-0.0376, +0.0139]	[-0.0466, +0.0192]
EEGNet L1	-0.0001	[-0.0040, +0.0047]	[-0.0060, +0.0059]
EEGNet L2	+0.0003	[-0.0028, +0.0040]	[-0.0042, +0.0048]

803

804 N.4 Resource effects and statistical limits

805 For the EEG-free prior, participant-dose changes from (0,0) to (8,0) are +0.2036
806 [+0.0955, +0.3116] on Emo and +0.1564 [+0.0492, +0.2637] on Urban. Stimulus-dose changes
807 to (0,8) are +0.4907 [+0.3581, +0.6233] and +0.3408 [+0.2161, +0.4655]. Joint changes to (8,8)

Table 25: EmoEEG-MC: uniform-grid EC-PC MAE. Paired pointwise intervals condition on fixed fitted predictions; t-normal is a heuristic sensitivity analysis.

Setting	Mean	Percentile interval	t-normal interval
Band-power R1	+0.0047	[-0.0004, +0.0099]	[-0.0019, +0.0113]
Band-power R2	+0.0039	[0.0000, +0.0078]	[-0.0013, +0.0090]
LaBraM R1	+0.0151	[+0.0061, +0.0242]	[+0.0034, +0.0269]
LaBraM R2	+0.0065	[+0.0016, +0.0112]	[+0.0001, +0.0129]
EEGNet L1	+0.0002	[-0.0008, +0.0013]	[-0.0012, +0.0016]
EEGNet L2	+0.0002	[-0.0004, +0.0009]	[-0.0007, +0.0011]

Table 26: Urban: uniform-grid EC-PC CCC. Paired pointwise intervals condition on fixed fitted predictions; t-normal is a heuristic sensitivity analysis.

Setting	Mean	Percentile interval	t-normal interval
Band-power R1	-0.0060	[-0.0432, +0.0376]	[-0.0519, +0.0398]
Band-power R2	-0.0069	[-0.0312, +0.0216]	[-0.0370, +0.0233]
LaBraM R1	-0.0173	[-0.0482, +0.0112]	[-0.0522, +0.0175]
LaBraM R2	-0.0035	[-0.0238, +0.0150]	[-0.0257, +0.0187]
EEGNet L1	+0.0073	[-0.0056, +0.0217]	[-0.0080, +0.0226]
EEGNet L2	+0.0086	[-0.0043, +0.0215]	[-0.0059, +0.0230]

are +0.6327 [+0.4810, +0.7844] and +0.4181 [+0.2928, +0.5435]. These are heuristic t-normal sensitivity intervals; the corresponding percentile intervals also exclude zero. They are not the same contrasts as $PC - P$ at a fixed stimulus dose in the main endpoint table.

The bootstrap reuses crossed participant and stimulus multiplicities across all fitted predictors. Averages are computed within block and then across blocks rather than pretending repeated grid predictions are new observations. The finite-cluster coverage diagnostics and invalid-draw counts constrain interpretation. These are fixed-fit uncertainty summaries, not repeated-training confidence statements or equivalence tests. Neither isolated pointwise intervals nor dose-family bands imply simultaneous coverage over every method, task, and secondary endpoint inspected during benchmark development.

N.5 Known-truth evaluator stress tests

We tested the estimator on outcomes with known construction before interpreting the EEG results. In the v6 oracle suite, 16 settings with 100 replicates each produced the following operating characteristics. “Detection” denotes a strictly positive interval and is not a universal power guarantee.

Table 27: Urban: uniform-grid EC-PC MAE. Paired pointwise intervals condition on fixed fitted predictions; t-normal is a heuristic sensitivity analysis.

Setting	Mean	Percentile interval	t-normal interval
Band-power R1	+0.0063	[-0.0012, +0.0144]	[-0.0024, +0.0151]
Band-power R2	+0.0037	[-0.0009, +0.0082]	[-0.0015, +0.0089]
LaBraM R1	+0.0091	[+0.0036, +0.0148]	[+0.0027, +0.0156]
LaBraM R2	+0.0038	[+0.0003, +0.0073]	[-0.0002, +0.0078]
EEGNet L1	+0.0027	[-0.0009, +0.0068]	[-0.0016, +0.0071]
EEGNet L2	+0.0012	[-0.0013, +0.0041]	[-0.0019, +0.0043]

Generator	Recorded behavior	Interpretation boundary
Labels only	mean difference exactly zero	algebraic negative control only
Independent noise	0/16 cells detected	finite evidence, not a proof of zero false positives
Trial signal	detection 0.85–1.00; coverage 0.92–1.00	detects the injected trial signal in these settings
Participant constant	minimum coverage 0.82	exposes finite-cluster weakness; no biological or causal interpretation

The v7 finite-cluster study planned 9,600 fits; 9,584 were valid and 16 were rejected by the declared support rule. In the target 32×24 cluster setting, percentile, basic, and t-normal coverages were 0.933, 0.930, and 0.957. Their minima in the deliberately small 8×6 setting were 0.837, 0.820, and 0.940. The often wider t-normal interval is therefore retained as a sensitivity analysis, not promoted as a generally calibrated solution.

N.6 Rule basis and three-state decisions

The checker implements observable evidence rules rather than inferring scientific validity from a score. Every rule returns allow, block, or unverifiable; the last state is preserved when required provenance is absent.

Rule family	Required evidence	Example boundary
Trial lineage	trial UIDs in every partition	repeated derived windows block trial-isolation claims
Identity holdout	participant/stimulus UIDs	overlap blocks only the identity claim it violates
Preprocess/selection	fit and selection UIDs	missing provenance is unverifiable, not a pass
Matched resources	predictor-specific label access	unequal calibration blocks an EEG-added-value claim
Fixed support	test-set hashes by comparator	unequal test support blocks a paired comparison

These rules operationalize documented leakage and benchmark-reporting concerns [3, 5, 15, 27]; they do not certify data truth, upstream pretraining history, or causal validity.

O Source-Linked Execution Cases and Reuse

The executed EEGain commit is d6892f5586181f345969651a9baedabd828bb1c5; its EEGDataLoader.split_train_val is in eegain/data/loader.py. LibEER is pinned to dddff9776dbdae21195fe320dff0a5ba61628a18; we execute get_split_index, index_to_data, and the empty-validation branch at lines 100–102 of LibEER/EEGNet_train.py. File hashes, selected source ranges, case origin, and outputs are recorded in external_audit/public_summary.json and its paired case table. Marker windows are generated from real trial identities, not independent EEG measurements. No external classifier score is inferred from the branch tests.

The same EEGain execution record produces different answers to different contracts: its held-out test participants remain excluded, while an additional requirement that validation trials be disjoint from training can be violated. Changing the claim to what was actually isolated is one valid repair. When a strict validation-trial contract is required, partitioning complete trials before generating windows is the appropriate construction instead. LibEER’s explicit train/validation/test path passes the exercised split checks; its empty-validation fallback should not be treated as independent validation. The machine-readable repair record submission_closure_v11/external_claim_repair.json verifies this strict-block/scoped-allow pair across all five exercised EEGain folds while holding the pinned function, trial universe, and execution output fixed. The repair changes the supportable claim; it does not alter or re-estimate a model score.

```
python scripts/run_workflow_contract_toy.py toy_contract
python -m openaffect_eeg.workflow_audit \
  --trials toy_contract/trials.tsv \
  --execution toy_contract/unmatched_calibration_execution.json \
  --contract toy_contract/unmatched_calibration_contract.json \
  --output toy_contract/checked.json
```

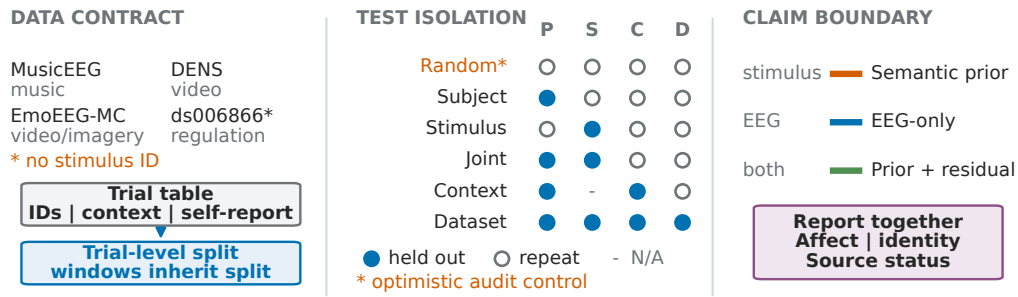


Figure S6: Original trial-first data and evaluation contract. Stimulus identity is unavailable for ds006866. The later Urban source and resource-matched predictor pairs are specified in the main methods and latest-evidence appendix.

860 The mismatch example exits with the documented blocked status; the matched example passes
861 observable checks. Omitting preprocessing evidence preserves uncertainty. This is a developer-
862 operated executable example, not independent human acceptance. The checker validates consistency
863 of supplied execution records, not adversarially falsified provenance.

864 **AI-assisted development.** AI assistance was used for protocol and code drafting, debugging,
865 literature triage, analysis support, and manuscript development. The recorded EEG predictions,
866 bootstrap summaries, and contract decisions are generated by the documented numerical models
867 and deterministic checks, not by an LLM acting as an emotion predictor or audit judge. Automated
868 checks and agent-run examples are not independent human review. Authors remain responsible for
869 source verification, scientific interpretation, and submission approval.

870 P Historical Overview Figures

871 The following approved figures retain earlier evidence without changing its source version. The initial
872 data layer covers the original four datasets; Urban is the later source described above. These figures
873 are not the latest matched-increment tables and must not be read as latent-emotion decompositions.

Identity-exposure response surfaces

Same fixed joint test trials in every cell; higher doses use labelled deployment resources

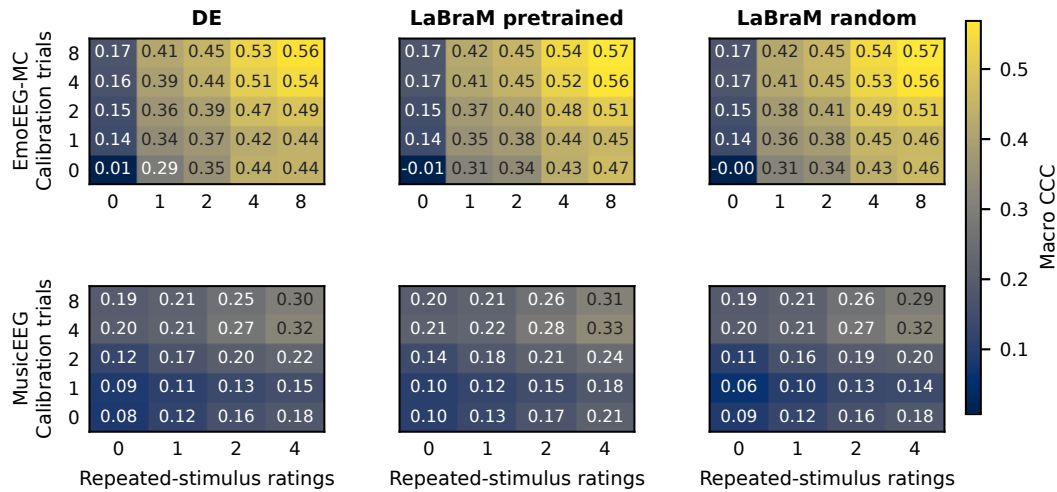


Figure S7: Original EmoEEG-MC and MusicEEG total calibrated prediction surfaces on fixed trials. This is total system CCC, not EEG added value. The new coverage analysis and matched contrasts are reported separately in the main tables; MusicEEG retains its original feasible grid rather than being relabelled as a full maximum-dose grid.

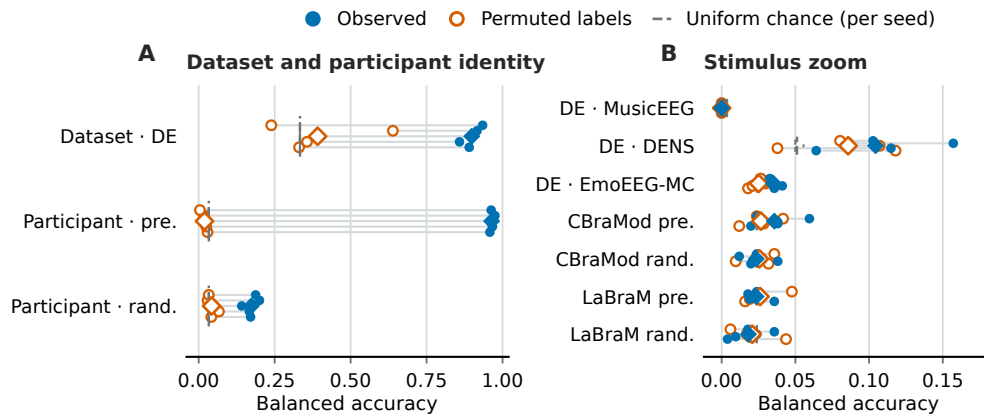


Figure S8: Original representation probes. Observed and permuted identity decoding describe available information, not proof that an affect predictor uses identity or that identity is a pure nuisance.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: The abstract and Introduction state the benchmark contributions, identity assumptions, and protocol-dependent scope; the Results and Limitations preserve those same boundaries.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

884 Answer: [\[Yes\]](#)

885 Justification: The Discussion contains a dedicated Limitations paragraph, with further

886 source-metadata, compute-accounting, and licensing limitations in the appendix.

887 **3. Theory assumptions and proofs**

888 Question: For each theoretical result, does the paper provide the full set of assumptions and

889 a complete (and correct) proof?

890 Answer: [\[N/A\]](#)

891 Justification: Section 3 defines resource-conditioned score contrasts and their arithmetic

892 identity, not a theorem about latent emotion or a causal decomposition.

893 **4. Experimental result reproducibility**

894 Question: Does the paper fully disclose all the information needed to reproduce the main ex-

895 perimental results of the paper to the extent that it affects the main claims and/or conclusions

896 of the paper (regardless of whether the code and data are provided or not)?

897 Answer: [\[Yes\]](#)

898 Justification: Sections 3–5, the historical appendices, and the latest-evidence appendix

899 specify sources, trial construction, resource budgets, estimators, paired statistics, and the

900 recomputation commands.

901 **5. Open access to data and code**

902 Question: Does the paper provide open access to the data and code, with sufficient instruc-

903 tions to faithfully reproduce the main experimental results, as described in supplemental

904 material?

905 Answer: [\[No\]](#)

906 Justification: Code, configurations, aggregates, metadata, and synthetic commands exist

907 in a versioned public release. The anonymous review endpoint is still incomplete, so this

908 working draft does not answer Yes. The answer can change only after actual anonymous

909 access is checked. Source recordings remain with their original providers.

910 **6. Experimental setting/details**

911 Question: Does the paper specify all the training and test details (e.g., data splits, hyperpa-

912 rameters, how they were chosen, type of optimizer) necessary to understand the results?

913 Answer: [\[Yes\]](#)

914 Justification: Section 4 gives the evaluation logic and model families; Appendices C–

915 F provide split ratios, adaptation details, validation rules, crops, patches, controls, and

916 statistical settings.

917 **7. Experiment statistical significance**

918 Question: Does the paper report error bars suitably and correctly defined or other appropriate

919 information about the statistical significance of the experiments?

920 Answer: [\[Yes\]](#)

921 Justification: Section 4.2 and the latest-evidence appendix define shared crossed-identity

922 paired intervals, invalid draws, fixed-fit uncertainty, and heuristic small-cluster sensitiv-

923 ity. Historical seed sign-flip tests are separately labelled and are not independent study

924 replication.

925 **8. Experiments compute resources**

926 Question: For each experiment, does the paper provide sufficient information on the com-

927 puter resources (type of compute workers, memory, time of execution) needed to reproduce

928 the experiments?

929 Answer: [\[No\]](#)

930 Justification: Appendix H reports CPU/GPU type, GPU memory, software versions, and

931 component boundaries, but exact wall-clock and energy totals were not logged consistently

932 for every preliminary and failed run.

933 **9. Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The work uses consented public research data under source terms, preserves submission anonymity, avoids re-identification, and releases no raw physiological recordings or restricted stimuli.

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Sections 6–7 discuss scoped evaluation benefits, privacy and identification risk, and why self-report prediction is not clinical or private-state access.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [Yes]

Justification: The release excludes raw EEG, source derivatives, restricted media, checkpoints, and real-identity linkage; identity probes are documented as risk measurements rather than identification tools.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: Appendix A and the latest-evidence appendix cite source versions and unresolved license caveats, including the Urban snapshot’s inconsistent declarations. Model cards record code, checkpoint provenance, and licensing boundaries.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The prepared release contains data/model/evaluation cards, Croissant metadata, protocol status, limitations, and reproducibility modes. Preparation is distinct from the still-incomplete anonymous hosting step.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: This work performs secondary analysis of public datasets and recruits no participants or crowd workers; collection procedures remain documented by the cited source studies.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

984 Justification: The authors conduct no new human-subject data collection. Ethical approvals
985 and consent for the source recordings are the responsibility of, and documented by, the cited
986 data providers.

987 **16. Declaration of LLM usage**

988 Question: Does the paper describe the usage of LLMs if it is an important, original, or
989 non-standard component of the core methods in this research? Note that if the LLM is used
990 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
991 scientific rigor, or originality of the research, declaration is not required.

992 Answer: [\[Yes\]](#)

993 Justification: The latest-evidence appendix discloses AI assistance in protocol and code
994 drafting, debugging, literature triage, and manuscript development. EEG predictions and
995 audit decisions are produced by the documented numerical models and deterministic code,
996 not by an LLM. Authors retain responsibility for verification before submission.